

## Prac-02

**Demonstrate how to build a Docker image using Ubuntu and execute an echo command when the container runs**

```
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\VICTUS> mkdir dockerfile_prep_1

Directory: C:\Users\VICTUS

Mode                LastWriteTime         Length Name
----                -
d-----         12-02-2026 12:31 PM             dockerfile_prep_1

PS C:\Users\VICTUS> cd dockerfile_prep_1
PS C:\Users\VICTUS\dockerfile_prep_1> notepad dockerfile
PS C:\Users\VICTUS\dockerfile_prep_1> docker build -t ub1 .
[+] Building 0.2s (1/1) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.1s
=> => transferring dockerfile: 2B                                0.0s
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory
PS C:\Users\VICTUS\dockerfile_prep_1> ren dockerfile.txt dockerfile
PS C:\Users\VICTUS\dockerfile_prep_1> dir

RAM 0.85 GB  CPU 0.08%  Disk: 5.01 GB used (limit 1006.85 GB)
```

**Step1:** Create a directory

**Step2:** Create a file named as dockerfile using notepad/nano

**Step3:** Paste the content given below

```
# Base image
FROM ubuntu:latest

# Metadata
LABEL maintainer="Student Demo"

# Avoid interactive prompts
ENV DEBIAN_FRONTEND=noninteractive

# Install required packages
RUN apt update && apt install -y php nano && rm -rf /var/lib/apt/lists/*

# Print message when container runs
CMD ["echo", "Hello Students"]
```

**Step4:** Rename the txt the file in dockerfile

```
Directory: C:\Users\VICTUS\dockerfile_prep_1

Mode                LastWriteTime         Length Name
----                -
-a----             12-02-2026  12:36 PM           223 dockerfile

PS C:\Users\VICTUS\dockerfile_prep_1> docker build -t ub1 .
[+] Building 155.2s (6/6) FINISHED
=> [internal] load build definition from dockerfile                                docker:desktop-linux 0.0s
=> => transferring dockerfile: 262B                                              0.0s
=> [internal] load metadata for docker.io/library/ubuntu:latest                 0.2s
=> [internal] load .dockerignore                                                 0.0s
=> => transferring context: 2B                                                  0.0s
=> [1/2] FROM docker.io/library/ubuntu:latest@sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b 0.1s
=> => resolve docker.io/library/ubuntu:latest@sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b 0.0s
httpd:latest      dd178595edd6      177MB      47.6MB      U
kavya4619n/ap1:v1.1 25109184c71b      13.1MB      3.95MB      U
kavya4619n/ap2:v2.1 69eae0896851      33.6MB      10.1MB      U
mongo:latest      6f55a078e025      1.27GB      333MB       U
mysql:latest      6b18d01fb632      1.27GB      283MB       U
nginx:latest      c881927c4077      240MB       65.7MB      U

RAM 0.86 GB CPU 0.17% Disk: 5.01 GB used (limit 1006.85 GB)
```

```
httpd:latest      dd178595edd6      177MB      47.6MB      U
kavya4619n/ap1:v1.1 25109184c71b      13.1MB      3.95MB      U
kavya4619n/ap2:v2.1 69eae0896851      33.6MB      10.1MB      U
mongo:latest      6f55a078e025      1.27GB      333MB       U
mysql:latest      6b18d01fb632      1.27GB      283MB       U
nginx:latest      c881927c4077      240MB       65.7MB      U
ub1:latest        a423edda30b1      412MB       115MB       U
ubuntu:latest     cd1dba651b30      119MB       31.7MB      U
PS C:\Users\VICTUS\dockerfile_prep_1> docker run -it ub1:latest
Hello Students
PS C:\Users\VICTUS\dockerfile_prep_1>
```

**Step5:** Build the image

**Step6:** Get inside the container

**Step7:** Verify the php installation